

ITALIAN SCHOOL MEALS OBSERVATORY

Pilot 2026

A multi-municipality pilot assessment of menu provision, transparency and school food-system governance

Technical Report v1.0 - public-document phase

Author: Vito Leone

Human Nutrition | Food Systems | Public Health Nutrition | Data & Evidence

7	170	12	17/17
Municipalities	Scheduled lunches coded	Governance domains	QA sampled rows concordant after correction

Version note

This v1.0 report documents the public-document phase of the pilot. Targeted access-to-information requests were initiated subsequently to obtain additional outcome data on satisfaction, food waste, non-conformities, procurement performance and related monitoring outputs. These future records are not yet incorporated here.

Executive summary

The Italian School Meals Observatory - Pilot 2026 is an independent applied food-systems project designed to test whether publicly available municipal information can be transformed into a structured, reproducible framework for comparing school-meal provision and governance. The pilot focuses on seven purposively selected Italian municipalities: Torino, Firenze, Napoli, Roma, Guspini, Cittadella and Argenta.

The project combines day-level menu coding with a standardized governance review. In total, 170 scheduled school lunches were coded using a frozen operational codebook. A 12-domain Governance Matrix was used to capture information on menu documentation, monitoring, stakeholder participation, special diets, procurement, sustainability, food waste, complaints, inspections and outcome transparency.

The pilot does not produce a municipal quality ranking. Its central purpose is descriptive: to identify what can be reliably observed from public documents, where municipalities appear convergent or divergent, and where the evidence base remains incomplete.

Across the seven cases, menu rotation for fish and legumes was relatively convergent, while governance documentation and outcome transparency varied substantially. Public administrations often publish detailed service specifications and procurement requirements, but far fewer quantitative outcome indicators - such as plate waste, dish acceptance, non-conformities, penalties or achieved sustainable-procurement shares - are readily available.

A stratified QA audit sampled 17 of the 170 coded rows. During this process, a source-version discrepancy was detected for Roma. Rather than patching the sampled row, all 45 Roma rows were re-coded against the updated official menu applicable to the observed framework. Post-correction sampled binary classification agreement was 17/17. The codebook itself was not changed; the intervention was a provenance and source-control correction.

1. Purpose and research questions

The pilot was developed as a portfolio-scale research and monitoring exercise at the intersection of public health nutrition, school food, data quality and food-system governance. It asks four practical questions:

- Can heterogeneous municipal school-meal documents be converted into a common analytical structure?
- What aspects of scheduled meal provision can be compared reproducibly across municipalities?
- Which governance and monitoring features are visible in public documentation, and which remain opaque?
- Where are the main gaps between documented service requirements and publicly available implementation outcomes?

The project is explicitly exploratory and purposive. It is not nationally representative and it does not infer actual individual dietary intake from published menus.

2. Study scope and case selection

Seven municipalities were included to create a heterogeneous pilot set spanning large metropolitan systems, medium-sized municipalities and smaller local authorities. The unit of menu analysis is one scheduled school lunch day.

Municipality	Coded lunches
Torino	20
Firenze	20
Napoli	20
Roma	45
Guspini	20
Cittadella	25
Argenta	20

Total coded scheduled lunches: 170.

3. Menu-coding methodology

The codebook v1.0 was frozen before final comparative analysis. Core rules were applied consistently across municipalities:

- One row represents one scheduled school lunch day.
- Cyclic menus of up to nine weeks are coded across the full cycle; long annual calendars are sampled using four sentinel weeks distributed through the year.
- Season-specific menus are treated within their applicable season unless multiple seasons are explicitly compared.
- Prospective tender menus are kept separate from observed provision.
- Unknown or unavailable public information is coded as blank/NA rather than zero.
- Fresh fruit excludes juice, yogurt, pudding, cake, gelato and fruit puree/polpa.
- Potatoes are not counted as a vegetable.
- Fish, fresh meat, cheese and eggs generally count when they are the designated main/second course, not merely incidental ingredients in a first course.
- Wholegrain is coded only when explicitly described as integrale/wholegrain.
- A plant main is coded only when a plant-protein item is clearly designated as the main course.
- Micronutrient adequacy is not inferred from dish names alone.

4. Governance framework

A standardized Governance Matrix was used to avoid reducing school-meal assessment to menu composition alone. Twelve domains were reviewed across the seven cases, covering the following dimensions:

- Menu and recipe documentation
- Nutritional validation / institutional oversight
- Portion and grammage information
- Special diets and accessibility

- Monitoring and inspections
- Stakeholder participation / meal commissions
- Satisfaction and acceptability monitoring
- Food waste measurement and prevention
- Non-conformities, penalties and corrective actions
- Sustainable procurement requirements
- Achieved procurement performance
- Public outcome transparency

A key analytical rule is that procurement requirements are not treated as achieved performance. Likewise, the availability of documentation is analyzed as a distinct dimension and is not equated with underlying service quality.

5. Quality assurance and provenance control

A stratified random QA audit was conducted on 17 of the 170 rows (10%), using fixed seed 20260920 and ensuring representation of all seven municipalities. The audit checked coding against the underlying public source material.

The audit identified a Roma source-version discrepancy. To protect dataset integrity, the entire 45-row Roma subset was re-coded against the official updated 2024 winter menu applicable within the observed 2021-31 July 2026 framework. This was treated as a provenance correction, not a codebook revision. After correction, sampled binary classification agreement was 17/17 (100%).

This episode illustrates a central methodological principle of the project: source versioning and document provenance are part of analytical quality, not merely administrative details.

6. Cross-municipality menu findings

The pilot suggests relatively modest variation in some recurring menu components, particularly fish and legumes, compared with the wider variation observed in governance and transparency. Fish appears approximately weekly in many of the coded cycles, while legumes commonly appear around one to one-and-a-half times per week. These are descriptive observations from scheduled menus, not measures of actual consumption.

Roma post-QA menu summary

Indicator	Roma coded frequency
Vegetables	4.333 / week
Fresh fruit	4.444 / week
Legumes	1.222 / week
Fish	1.000 / week
Fresh meat	1.556 / week
Processed meat	0.444 / month
Wholegrain	0.556 / week
Eggs	1.222 / week
Fried/breaded items	~3.111 / month

These frequencies reflect the corrected coded schedule and should not be interpreted as intake, adherence or quality scores.

7. Municipal governance profiles

Torino

Torino provides extensive public documentation on menus, recipes and primary-school grammages, with ASL validation/endorsement and a formal quality-management context. Commissioni Mensa and the RisTò participation system are visible, alongside a substantial inspection framework.

The municipality reports more than 480 municipal inspections per year, including surprise visits, and publishes a 2025 primary-school satisfaction survey based on 3,435 valid questionnaires among 15,369 service users (~22.4% response). Reported mean scores include overall satisfaction 4.84/10, menu variety 5.40, balance 5.49, foods liked 4.40, fish 4.09, legumes 5.00, vegetables 4.94, fruit 5.90, bread 6.41, dish temperature 4.75, overall dish liking 4.28 and staff courtesy 6.71. These results are respondent outcomes and should not be generalized to non-respondents.

Publicly visible gaps remain for quantitative plate waste, aggregate sanctions/non-conformities and achieved procurement shares.

Firenze

Firenze presents comparatively strong governance and outcome documentation. The pilot identified structured monitoring of daily gradimento, Commissioni Mensa activity, child interviews, administrative outcome information and food-waste initiatives. A 41-week calendar was handled through four sentinel weeks. Quantitative waste and procurement-performance data remain less complete than process documentation.

Napoli

Napoli operates a large school-meal framework with detailed technical documentation, recipes and raw grammages by school level, including primary school. Governance includes ASL Napoli 1 Centro involvement, three levels of Refezione Commissions and daily or external control mechanisms. Public 2025 review material describes satisfaction qualitatively as sufficient and external checks as excellent, while also referring to sanctions and corrective action; however, the underlying numeric datasets were not publicly located in the pilot phase.

Procurement documentation includes CAM, organic, km0/short-chain and DOP/IGP requirements. Waste-prevention processes are documented, but quantitative waste outcomes were not identified. A March 2026 menu declares fruit and wholegrain bread without mapping them to individual days, so those day-level variables remain unknown for that source.

Roma

Roma is the largest system in the pilot, with approximately 150,000 daily users, more than 30 million meals per year and around 730 schools. The observed framework includes seasonal menu cycles, daily technical inspections by qualified professionals, formal Commissione Mensa structures, special diets and income-related relief.

A 2024 citywide 23-question consultation on school meals was identified, but detailed results were not located in the public-document phase. A new 48-month, 15-lot tender of approximately EUR 766 million was in evaluation in September 2026; its prospective menu and sourcing criteria are therefore not treated as observed performance in this report.

Guspini

Guspini stands out for detailed public portions and kcal information by school level, ASL-approved seasonal menus and explicit objectives related to nutrition, gradimento, special diets, organic sourcing, short supply chains and waste reduction. Historical evidence includes a meal-commission framework and inspection activity, while a 2025 SWITCH initiative highlights BIO/km0 local supply themes. Historical data are kept separate from current performance. Current quantitative outcomes remain limited.

Cittadella

Cittadella provides a particularly explicit contractual framework for the 2025/26-2029/30 period. The service requires satisfaction/performance measurement, Commissione Mensa activity, direct measurement of food prepared but not distributed and plate leftovers, an anti-waste project, semiannual CAM sourcing/waste reports and a penalty system.

The tender also specifies category-level sustainable sourcing requirements, including 100% organic yogurt, eggs and juice, at least 70% organic shares for major plant-food groups and sustainable fish criteria. The analytical gap is not absence of requirements, but limited public availability of the resulting performance reports.

Argenta

Argenta combines a formal parent meal committee with economic and service-coverage information. Public 2024 administrative data indicate 14 schools, 1,619 potential pupils, 1,046 pupil users, 726 primary pupils and 152,289 meals, corresponding to 64.6% potential-demand coverage. Reported service expenditure was EUR 900,815.11, total revenue EUR 816,436.88 and user-fee revenue EUR 743,872.88, with total cost coverage of 90.63% and user-fee coverage of 82.58%.

A 2024 school-meal satisfaction survey is explicitly documented as having been shared with the provider and committee, but detailed results were not located. The municipality also applies income-related reductions and sibling discounts and has planned a new 312-pupil canteen investment.

8. Main analytical interpretation

The most important cross-case observation is that municipalities often disclose far more information about service design than about service outcomes. Menus, procurement specifications, eligibility rules and monitoring procedures are commonly visible. By contrast, plate waste, dish acceptance, detailed customer-satisfaction results, non-conformity aggregates, penalties, corrective actions and achieved procurement shares are much less consistently published.

This creates a recurring evidence gap between what a service is required to do and what can be demonstrated publicly about what it actually achieved. The pilot therefore treats outcome transparency as a substantive food-system monitoring issue rather than a mere communications feature.

9. Limitations

- Purposive seven-municipality sample; findings are not nationally representative.
- Published menu provision is not equivalent to food served, food selected or food consumed.
- Municipal documents differ in date, granularity, terminology and publication practices.
- Unavailable public data are coded as unknown/NA rather than assumed absent.
- Historical information is not treated as evidence of current performance.
- Procurement requirements are not treated as achieved procurement shares.
- No municipal quality ranking is produced; any exploratory alignment metrics are methodological, not legal-compliance or service-quality scores.
- The report predates the outcome-data access requests sent to the seven municipalities and will require updating when those records are received.

10. Outputs produced

- QA-audited master workbook with 170 coded lunches.
- Municipality summary dataset and governance matrix.
- Frozen operational codebook v1.0.
- Stratified QA audit with documented Roma provenance correction.
- Technical Report v1.0.
- One-page Executive Summary.
- Reproducible analysis pack prepared for R, with CSV exports and analysis script.

- Targeted access-to-information framework for outcome-data retrieval.

11. Next phase

On 25 September 2026, targeted access-to-information requests were sent to all seven municipalities. The requests seek already existing records on satisfaction/gradimento, inspections, non-conformities and penalties, food waste, procurement performance, meal-commission outputs and, where available, special-diet utilization. Napoli was submitted through the municipal SmartURP workflow; the other requests were submitted through the relevant FOIA/PEC channels.

The planned v2.0 report will integrate any newly received outcome data while preserving the distinction between published menu schedules, contractual requirements, monitoring processes and observed outcomes.

12. Positioning of the project

The Observatory is designed as an applied evidence project rather than a purely academic exercise. It demonstrates an end-to-end workflow spanning research design, public-document review, coding, data cleaning, quality assurance, governance analysis, evidence synthesis and targeted data acquisition. Its intended professional relevance lies in public health nutrition, school food, food-systems monitoring, programme evaluation and evidence translation.

Contact

Author: Vito Leone

Professional focus: Human Nutrition | Food Systems | Public Health Nutrition | Data & Evidence

Email: vitoleone.nutrizione@gmail.com